# Theoretical minimum: Paper 008

**AuthorityEffect:** None. Read the fictional Apartment 50 preface, then attempt these questions before opening the worked study guide.

These eight learning rungs, L0 to L7, are a teaching sequence. They are separate from MSS stage numbers and the evidence grades NONE, SYNTAX and COMPUTED. Use exact integers or rational fractions. A graph edge is a declared constraint; no filled face is implicit.

Each question earns two points: one for the calculation or concrete disposition, and one for its evidence boundary. The total of 32 is a self-study score. Record the failed obligation even when the arithmetic is correct.

## L0: Packets and admission

**Q01.** Using the archived actual parser, complete a table with three columns: parses, active at MSS-0, active at MSS-1. Test `<>`, `<0>`, `<2>` and `op<<1><0>>`. Explain why `op` does not enlarge the active data alphabet.

**Q02.** In the actual maps, calculate `d_1(u_0(parse("<>")))`, then predict `u_0(parse("<0>"))` and `d_1(parse("<1>"))`. What information does the refusal preserve? Why is the historical toy's `None` a different interface?

## L1: What the round trip proves

**Q03.** An illustrative old carrier is `{a,b}`; its lift has image `{ua,ub}` inside `{ua,ub,x}`. Let `d(ua)=a`, `d(ub)=b`, and `d(x)=NON_DESCENT`. Prove injectivity of the lift from the round-trip equation. Does $u(d(x)) = x$ follow?

**Q04.** A new proof system keeps every old term unchanged and adds an axiom asserting an old-language sentence $P$ that the old system cannot prove. Does the syntactic round trip survive? Does conservativity of old-language provability survive? Name the missing kind of theorem.

## L2: Namespaces and grades

**Q05.** For stages $N \geq 22$, $b(N) = 128 + 8(N - 22)$. Calculate the eight-wide blocks at stages 22, 23 and 100. Prove adjacent blocks are disjoint. If all opcodes must fit an unsigned eight-bit wire, what is the last permitted stage? Does any calculation here implement that stage's semantics?

**Q06.** A packet passes its own-stage admission and fails one-stage-earlier admission. A proposed checker accepts its positive fixture **and also accepts its required negative fixture**. Which syntax evidence is available, and why is the COMPUTED obligation unmet? Can an unrelated Lean file or a level number fill that gap?

## L3: Every certificate clause

**Q07.** On the directed cycle $0 \to 1 \to 2 \to 0$, use loads $c = (1, 0, 0)$, unit capacities and zero vertex potentials. For flows $y = (1, 0, 0)$, compute primal cost, $c^{\mathsf{T}} y$, gap, and divergence (inflow minus outflow). Is the complete certificate valid? Repeat the balance check for $y = (1, 1, 1)$.

**Q08.** Glue dictionaries on $\{a, b\}$ and $\{b, c\}$: the first assigns $a = 0, b = 1$, and the second assigns $b = 1, c = 0$. Then change the second value at $b$ to zero. Which case glues? What does this finite check leave unproved about the file title "Sheaf Descent"?

## L4: Bind the instance

**Q09.** Edge indices 0, 1, 2 are $A \to B = 2$, $B \to C = 3$, $A \to C = 6$. Set $\pi(A) = 0$. Produce either a valid potential or a closed signed walk that proves contradiction. Check its edge continuity, closure and signed claim sum.

**Q10.** Change only the third claim to 5 and give the new certificate. A different proposal instead replaces all claims by zero and gets a zero defect. Which original constraints did it preserve? State what an acceptance rule must bind before "accepted" can mean repair of the requested instance.

## L5: Read the corpus and the statistics

**Q11.** At the census commit, 40 challenges are delivered solver packages with two checker modules, 2 are model-measurement specifications, and 8 are specification-only. Check the total and the disjoint partition. Does folder membership certify execution, correctness or a completed model measurement? What does a missing scoring receipt imply?

**Q12.** In the three-arm C005 run, Arm 1 has 25 correct cases; discordant counts against Arm 2 are $b = 6, c = 17$. Calculate Arm 2's count and the exact two-sided binomial McNemar value. The separate hybrid run has Arm 2 at 38/50. Is 37/50 an observed result? Which protocol and data limitations qualify these tables?

## L6: Scoring and custody

**Q13.** The eight K4 records have one trace per model/condition cell. The scratchpad and hinted prompts contain the three repair values; the other two conditions do not. What does an assisted success establish? Give two changes required for a defensible unassisted comparison.

**Q14.** A response says `VERDICT: NOT CONSISTENT`. Another contains conflicting explicit verdict lines. State their score disposition. Two well-formed responses have the same verdict but different raw text: should their new receipt digests match? Can a public unkeyed hash authenticate the provider?

## L7: Resolve and replay honestly

**Q15.** Historical mode v1 requests an exact revision and path that are unavailable, while active mode v2 has a file of the same name. Give the v1 disposition. If no optional SHA-256 was supplied, which extra comparison was not performed? Does byte recovery settle the cited mathematical claim?

**Q16.** A Lean process exits 0 but emits no declaration census. May the named verification gate pass? Distinguish the 39 selected theorem names, the retained 576 package-local environment declarations, and the separate 52 Level Engine payloads. If this host cannot start Lean, what may its review record honestly report?

## Evidence worksheet

For each answer retain: the declared input, your prediction, the calculation or named check, the result, and the remaining obligation. A malformed contract calls for refusal; missing execution is not a PASS. The study guide supplies worked answers, common mistakes and offline commands.